

Transfer Settlement Network

IDENTITY-FIRST · INTENT-BASED · PRIVACY-PRESERVING SETTLEMENT

Turning wallet-address payments into identity-based,
cryptographically authorized, privacy-bound settlement

Solana debit



TSN Node



Creditcoin Hub



Settlement

Built with Creditcoin & the Attestcoin Protocol · Track: DeFi (Payments & Settlement)

THE PROBLEM

Payments are fragmented. Settlement shouldn't be.

Crypto made value transferable, but payments remain fragmented. token standards, fees, and transaction mechanisms differ across chains. there is no unified identity, authorization policy, privacy boundary, liability accounting.

A message can describe a payment — but it is not the payment value. needs a real liquidity source, verified, with replay protection and privacy.

Fragmented

Wallet formats, tokens, fees & rails differ per chain

No identity layer

Payments expose wallet addresses, not portable identity

No privacy boundary

Sender, recipient & balance metadata leak on-chain

No unified liability

No replay protection or liability accounting standard

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Bank account number

→ TIN — Transfer Identity

One portable identity

One TIN, authorized settlement addresses across chains

Payment network

→ TSN coordination layer

Creditcoin-first settlement

Solana debits; funded Creditcoin liquidity settles

Institution-to-institution instruction

→ Cryptographically authorized

Settlement network

→ Solana, Creditcoin or another supported rail

Institutional controls

→ On-chain authorization

No oracle, no admin

Attestcoin proves cross-chain facts used for settlement

HOW IT WORKS

The Creditcoin-first settlement flow

1. Intent

User creates & signs a TIN debit intent on Solana.



2. Route check

Attestcoin SDK verifies destination liquidity via Creditcoin ASC route check.

3. Commit

TSN Node authorizes; Cranker Job 1 submits; Solana program creates the debit commitment.

4. Validate

TSN Node reads & validates the commitment, signs the Creditcoin exit authorization.

5. Settle

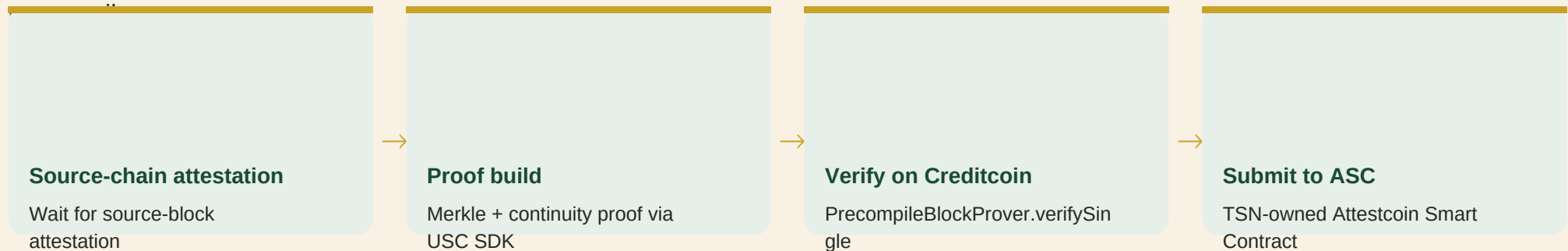
Cranker Job 2 submits;
CreditcoinSettlementHub consumes it —
replay protection + liquidity check.

6. Payout

Creditcoin pays recipient & fee
recipient in one transaction.

Attestcoin Protocol, used as a core feature

TSN deploys a TSN-owned Attestcoin Smart Contract on Creditcoin. The contract verifies proof-backed liquidity events from an approved destination EVM route through Creditcoin's native Block Prover



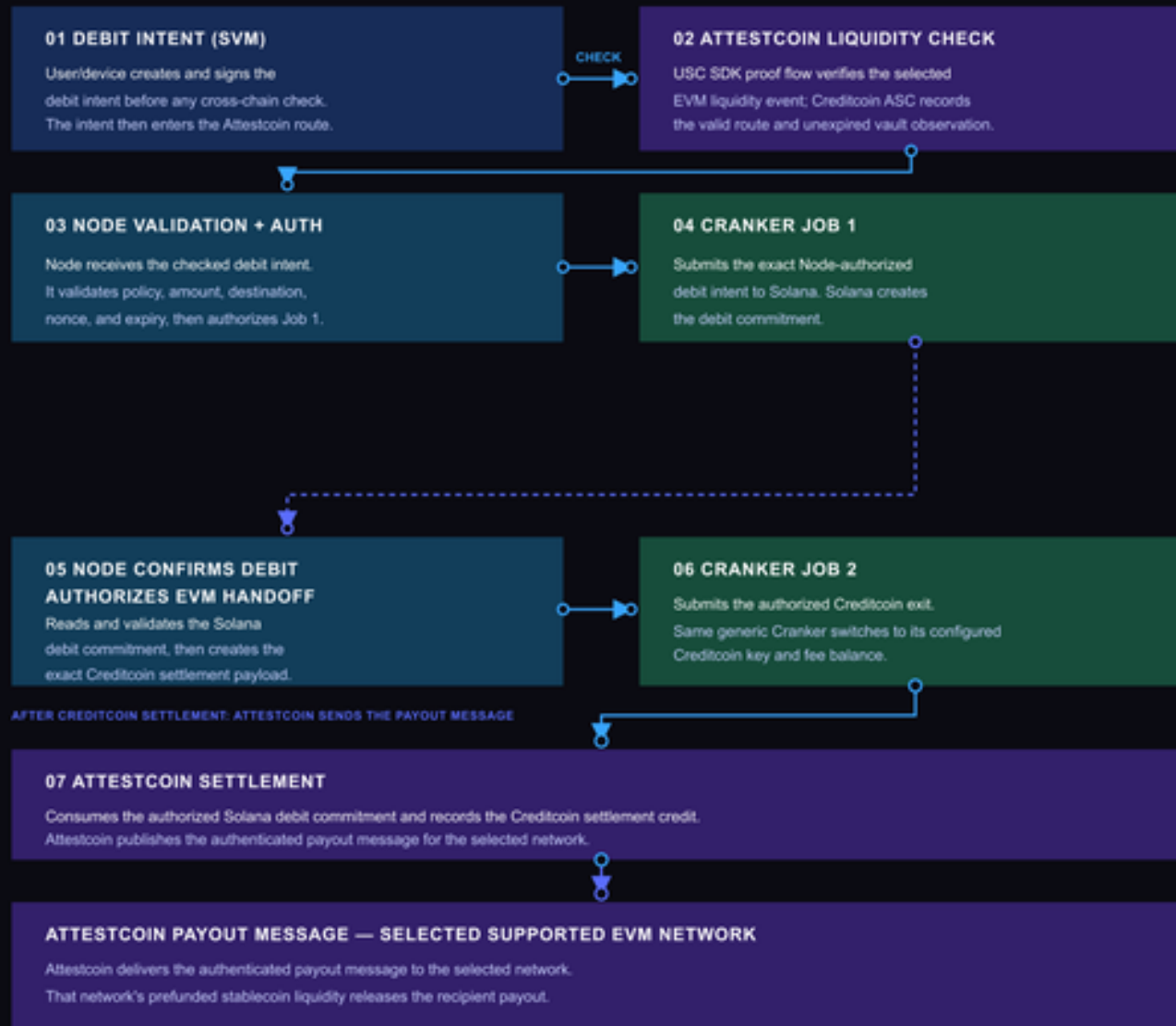
The ASC checks before recording a verified observation:

- Registered source chain & liquidity emitter
- Route & token match the destination payout vault
- Amount within available stablecoin liquidity
- Nonce not consumed (replay protection) & not expired
- LiquidityAvailable event: route · token · amount · nonce · expiry

DEBIT → EXIT CROSS-CHAIN FLOW

SVM → EVM, powered by Creditcoin settlement.

DEBIT INTENT → ATTESTCOIN CROSS-CHAIN CHECK → NODE AUTHORIZATION



From the monorepo

- `tsn-crosschain/contracts` —
`CreditcoinSettlementHub.sol`,
`DestinationLiquidityASC.sol`, `DestinationLiquidityRegistry.sol`,
`SepoliaAnchor.sol`,
`TSNERCLiquidityVault.sol`,
`TSNSettlementExecutor.sol`,
`TinExitAttestedASC.sol`
- `worker/` — `destination-liquidity-attest-worker.ts`,
`tin-exit-attest-worker.ts`, `usc-proof-helper.ts`
- `tsn-cranker-sdk` & `tsn-cranker-op-daemon/scripts/cranker.ts` —
generic multi-network submitter
- `tsn-node`, RPC gateway, Solana program workspace — `identity`,
authorization, epoch treasury, opaque slot settlement
- TCap — encrypted private balance snapshots & reservation

Live links

github.com/Trustlink-Labs/TSN-Protocol

trustlink-pay.vercel.app · trust-link-tsn.mintlify.com

LIVE PRODUCT TrustLink Pay — the product on TSN

TrustLink Pay shows TSN in action: private, identity-first, confidential crypto payments on Solana.

- Pay with a phone number or 10-digit TIN — not a wallet address
- Identity layer (TIP) + purpose-bound receiving identities (ZK-PRU)
- Private balances via TCap encrypted snapshots; opaque slot settlement
- Epoch treasury funding; Cranker execution; cryptographic verification

Live on Solana Devnet

trustlink-pay.vercel.app

Docs: trust-link-tsn.mintlify.site · Blog: tsn-protocol.blogspot.com

The screenshot displays the TrustLink Pay website interface. At the top, there's a navigation bar with links: Protocol, Identity, Attestations, Settlement, Developers, OPERATOR, and OPEN DAPP. The main heading reads "Private, Identity-first, and Confidential Crypto Payments on Solana". Below this, a paragraph explains that TrustLink Pay is built on the Transfer Settlement Network (TSN) as a decentralized settlement layer for Solana, turning payment intents into private, cryptographically verifiable execution through epoch funding, Cranker execution, and opaque slot settlement. It mentions that users can pay using a phone number or 10-digit TIN instead of a wallet address, with private settlement and optional identity trust context. Two buttons are present: "Learn about TSN" and "Read the repo".

A diagram titled "HOW A PAYMENT WORKS?" illustrates the payment flow in six steps: 1. Alice enters Bob's Identity, 2. Intent enters Mempool, 3. Cranker submits on-chain, 4. Escrow locks the funds, 5. Private claim pays Bob, and 6. Proof settles at epoch. The diagram also shows a "SETTLEMENT LOCK" and a "CYCLE" indicator.

Below the diagram, four key components are listed: 1. IDENTITY LAYER (TIP) for payment identity and optional trust attestations, 2. PRIVACY LAYER (ZK-PRU) for purpose-bound protected receiving identities, 3. SETTLEMENT (TSN) for payment intents, epochs, Crankers, fees, and receipts, and 4. CONFIDENTIAL ASSETS (TCAP) for reserve metadata, commitments, nullifiers, and private roots.

At the bottom, a section titled "BLOCKCHAIN SOLUTION" explains why TrustLink Pay is a modern blockchain solution. It states that a useful blockchain solution should make payments easier while preserving verifiability. TrustLink Pay achieves this by combining identity-first payments, stablecoin-native flows, private settlement routing, and developer APIs. It notes that TrustLink Pay is the Web3 payment system users interact with, and TSN is the settlement protocol that moves payment work through Crankers, ZK-

NEXT Roadmap

Done

- Identity-first settlement live on Solana devnet
- Creditcoin-first cross-chain path implemented (BUIDL CTC)
- TSN-owned ASC + proof workers in repo

Next

- Additional supported EVM settlement rails (Base payout vault drafted)
- Mainnet funding & liquidity bootstrap on Creditcoin
- Independent cryptographic verification of Solana commitments

Later

- Wider Attestcoin message-route payout matrix
- Institutional controls & liability tooling
- Partner network integrations via CEIP

TEAM Built by TrustLink Labs



TrustLink Labs is an AI-native blockchain technology lab building open financial infrastructure. We design privacy, identity, and payment protocols that power the next generation of decentralized applications.

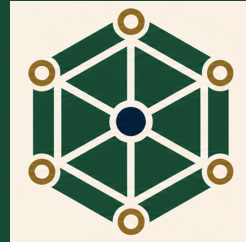
- TrustLink Pay — identity-based stablecoin payments built on TSN
- Open monorepo with SDKs, node services, cross-chain contracts & docs
- Owner: Big Dreams Web3 (1-member team — no blockers, full autonomy)

Links

github.com/Trustlink-Labs/TSN-Protocol

trustlink-pay.vercel.app · [@Trustlinklabs](https://twitter.com/Trustlinklabs) · tsn-protocol.blogspot.com

docs: trust-link-tsn.mintlify.site



Payments that settle like institutions — without the institutions.

TSN · TrustLink Pay · Built for the real world